GABRIELA SANCHEZ RODRIGUEZ

grodrguez3@gatech.edu ◇ linkedin.com/in/gabriela-sánchez-rodríguez

Machine learning researcher with experience developing deep learning models, statistical methods, and custom neural network architectures for complex, noisy, high-dimensional datasets. PhD researcher in bioengineering with an MS in Computer Science (Machine Learning), specializing in physics-informed deep learning, model optimization, and scientific inference. Experienced in designing rigorous computational experiments, extracting signal from noisy data, and translating research models into efficient software. Seeking an ML Research Scientist role applying these methods to forecasting and systematic decision-making.

EDUCATION

Georgia Institute of Technology — Atlanta, GA
PhD, Bioengineering (School of ECE) — Spring 2027
MS, Computer Science (Machine Learning) — 2023–2025

Universidad Carlos III de Madrid — Madrid, Spain
BS, Biomedical Engineering (Minor: Advanced Medical Imaging) — 2017–2021

SKILLS

Machine Learning	Deep Learning, Neural Network Architecture Design, Physics-Informed Neural Networks, Representation Learning, Model Optimization, Bayesian Optimization, Uncertainty Quantification, High-Dimensional Statistics, Dimensionality Reduction
Frameworks and Tools	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Weights & Biases
Programming	Python, R, MATLAB
Methods	Statistical Inference, Experimental Design, Optimization, Signal Extraction, Computer Vision, Scientific Computing
Data	High-Dimensional Data, Noisy Measurements, Sequential/Multitimestep Data, RNA-seq, Biomedical Imaging
Interests	Trail running, Quaternity Chess

RESEARCH EXPERIENCE

Georgia Institute of Technology — Aug 2021 – Present
Graduate Research Assistant; Atlanta, GA

Physics-Informed Neural Networks (DeepVFM)

- Designed a custom neural network architecture that learns optimal feature functions directly from data, replacing a hand-engineered statistical estimation method and automating what was previously a manual, expert-driven modeling step.
- Built a one-shot optimization framework that embeds physical boundary conditions as mathematical constraints directly in the training objective, cutting parameter identification from a multi-step classical pipeline to a single fast inference pass with **5% error**.
- Engineered a multi-timestep aggregation strategy that fuses sequential simulation snapshots into a single inference pass, improving robustness of learned physical parameters under noisy, high-dimensional measurements.
- Validated the framework across multiple material systems and loading conditions, demonstrating a scalable, generalizable alternative to system-specific classical inverse solvers.
- Ran systematic experiments across architecture variants, loss formulations, and noise regimes to characterize model robustness and generalization, establishing DeepVFM as a faster, more accurate alternative to classical inverse solvers.

Retinal Segmentation Software

- Trained U-Net and DenseNet architectures with cross-entropy loss to segment the retina, choroid, and sclera in rodent OCT images.
- Built and managed version-controlled pipelines to streamline model experimentation and Bayesian optimization workflows in Weights & Biases.
- Designed preprocessing pipelines to improve inference speed and postprocessing pipelines using classical image processing algorithms to correct segmentation errors.
- Built and distributed two standalone segmentation applications, the field's first tools for segmenting injured rodent eyes, achieving over **79% accuracy** even in diseased tissue.

High-Dimensional Statistical Analysis / RNA-seq Analysis

- Applied high-dimensional statistical inference methods, including hypothesis testing with multiple-comparison correction and dimensionality reduction, to isolate true signal from noise across four large-scale, high-dimensional datasets.
- Uncovered a novel, statistically significant pattern in high-dimensional biological data undetectable by univariate analysis, identifying the first molecular signature linking menopause to inflammatory and extracellular matrix pathway changes in the posterior eye.

SELECTED PUBLICATIONS

Full list on Google Scholar

1. **Sánchez-Rodríguez, G.**, Sampaio de Oliveira, M. L., Nguyen, T., & Feola, A. J. (*In prep.*) DeepVFM: Neural Networks Discovery of Virtual Fields for Stable Inverse Constitutive Identification of Hyperelastic Materials.
2. **Sanchez-Rodriguez, G.** et al. RetOCTNet: Deep Learning–Based Segmentation of OCT Images Following Retinal Ganglion Cell Injury. *TVST*, 14(2):4. (2025).
3. Ethier, C. R., Goyal, V., **Sánchez-Rodríguez, G.** et al. DL-based morphometric analysis of RGC axons in a rat model of glaucoma. *IOVS*, 62(8):2346. (2021).
4. Chopra, S., **Sanchez-Rodriguez, G.** et al. MedMoE: Modality-Specialized Mixture of Experts for Medical Vision-Language Understanding. **CVPR Workshop** MMFM-BIOMED. (2025).
5. **Sanchez, G.** et al. Deep-Learning segmentation of retina, choroid, and sclera from mice OCT scans. *IOVS*, 66(8):2173. (2025).
6. **Sanchez, G.** et al. DeepRetNet 2.0: Deep-Learning segmentation of retina, choroid, and sclera from rat OCT scans. *IOVS*, 65(7):2324. (2024).

SELECTED COURSEWORK

Deep Learning · Machine Learning · Mathematical Foundations of Machine Learning · Algorithm Design and Analysis · Uncertainty Quantification · Vision and Language Models · Advanced Computer Vision

SELECTED FELLOWSHIPS & RESEARCH AWARDS

Fellowships: La Caixa Fellowship (2022–2024) · P.E.O. International Scholar Award (2023, 2024)
Research Awards: CRIDC Poster Award (2026) · BrightFocus Foundation Travel Award (2021, 2025) · Grad Cohort Travel Award, CRA-WP (2022, 2023, 2024)